Algorithms and Computability

Lecture 4: Reductions

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Yesterday in Algorithms and Computability

We have seen

- Diagonalization
- and the non-computability of the halting problem.
- Also: closure properties of computable and computably-enumerable languages.

Encoding of Turing Machines

Let $M = (Q, \Sigma, \Gamma, s, t, r, \delta)$ be a DTM. We assume without loss of generality $Q = \{1, 11, \ldots, 1^{|Q|}\}$ and $\Sigma = \{0, 1\}$. Also, let $rep_\Gamma \colon \Gamma \to \{1, 11, \ldots, 1^{|\Gamma|}\}$ be an encoding of Γ such that $rep_\Gamma(0) = 1$, $rep_\Gamma(1) = 11$, and $rep_\Gamma(\sqcup) = 111$.

Then, M is encoded by the word $\ulcorner M \urcorner$ over $\{0, 1\}$ defined as follows:

$$1^{|Q|}\, 0\, 1^{|\Gamma|}\, 0\, s\, 0\, t\, 0\, r\, 0\, w_\delta$$

where w_δ is the list of encodings of transitions. Each $\delta(q, b) = (p, a, d)$ is encoded by

$$q\, 0\, rep_\Gamma(b)\, 0\, p\, 0\, rep_\Gamma(a)\, 0\, dir(d)\, 0$$

where $dir(-1) = 1$ and $dir(+1) = 11$.

The Halting Problem

- So, (the encoding of) a DTM can be the input for a DTM.
- Thus, we can formulate decision problems about DTMs!
- A particular interesting (both practically and theoretically) problem is the **halting problem**:

$$\mathrm{HP} = \{\langle \ulcorner M \urcorner, w \rangle \mid M \text{ is a DTM that halts on input } w\}$$

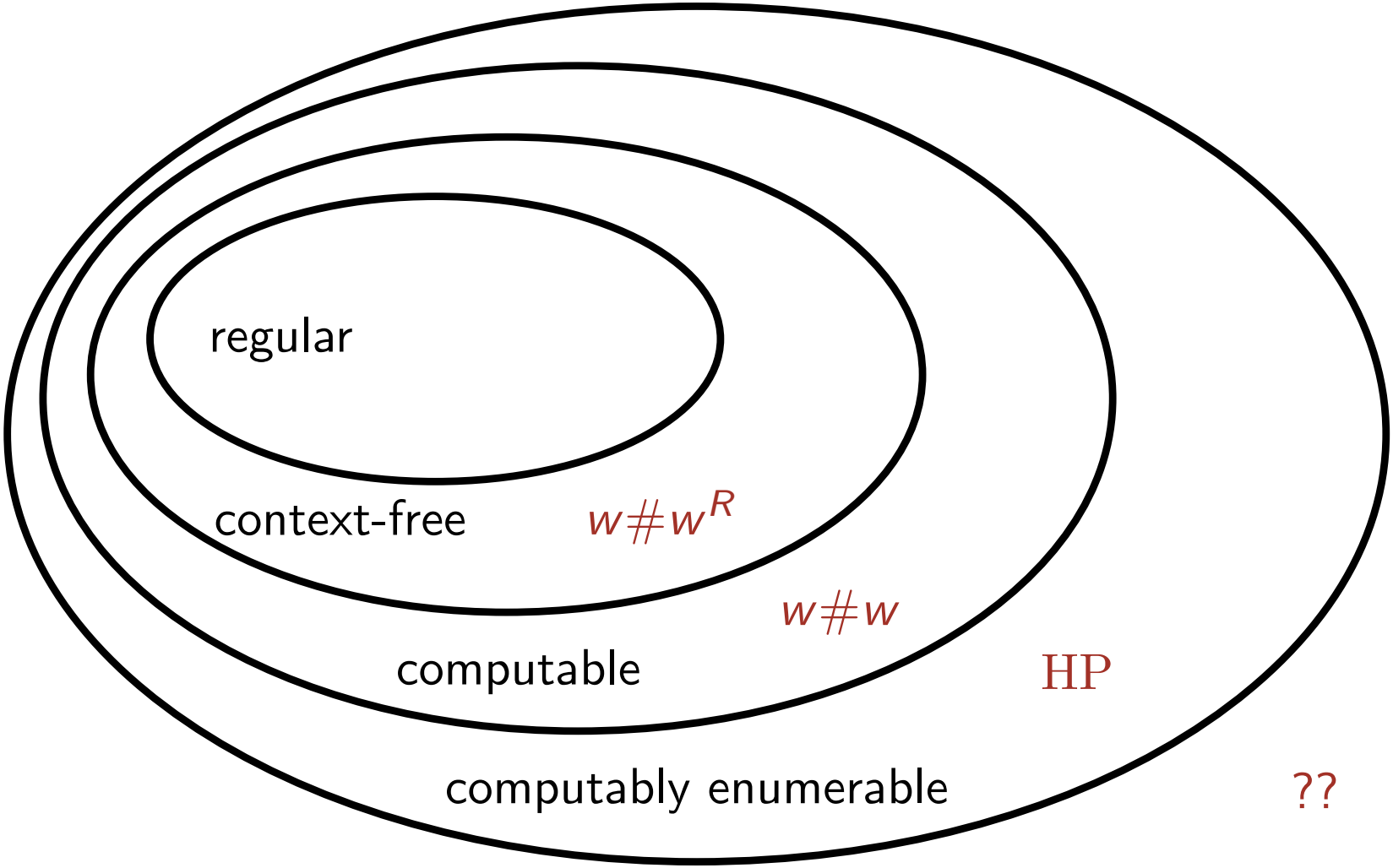
- Here, $\langle \cdot, \cdot \rangle$ is a function that encodes two words x, y over $\{0, 1\}$ by a single word $\langle x, y \rangle$ over $\{0, 1\}$. See the book (page 89) for details.

Theorem (Turing 1936)

The halting problem is not computable.

Today

- How to show more problems non-computable?
- How to show that a problem is not computably-enumerable?



Agenda

1. **Intuition**
2. Reductions
3. Applications
4. More Non-computable Problems

The Acceptance Problem for Turing Machines

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- If M rejects w, then M' accepts w.
- If M does not halt on w, then M' does not either.

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- Assume there is a halting DTM A for AP.
- Then, the following DTM H computes HP.
  1. Given input $\langle \ulcorner M \urcorner, w \rangle$, write $\langle f(\ulcorner M \urcorner), w \rangle$ on tape.
  2. Simulate A on that input, which will halt.
  3. If A accepts, H accepts as well.
  4. If A rejects, H rejects as well.
- This is a contradiction, HP is not computable. So AP cannot be computable.

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$$\langle \ulcorner M \urcorner, w \rangle \in \mathrm{AP} \quad \Leftrightarrow \quad \langle f(\ulcorner M \urcorner), w \rangle \in \mathrm{HP}$$

- There is a DTM H accepting HP (see Exercise sheet 3).
- The following DTM A accepts AP.
  1. Given input $\langle \ulcorner M \urcorner, w \rangle$, write $\langle f(\ulcorner M \urcorner), w \rangle$ on tape.
  2. Simulate H on that input.
  3. If H accepts, A accepts as well.
  4. If H rejects, A rejects as well.
- So, AP is computably-enumerable as well.

The Usefulness of Reductions

In general, we have reduced instances of a problem A to instances of another problem B (using a function).

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In the following, we formalize the notion of "reduction" and show some applications.

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In the following, we formalize the notion of "reduction" and show some applications.

Note

There are many different notions of reductions for different applications. We focus here on one notion that is useful for our goals.

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Caution

Let us "reduce" the halting problem

$$\mathrm{HP} = \{\langle \ulcorner M \urcorner, w \rangle \mid M \text{ is a DTM that halts on input } w\}$$

to the computable language $O = \{1\} \subseteq \mathbb{B}^*$ by defining the following function f:

- If $w \in \mathrm{HP}$, then define $f(w) = 1 \in O$.
- If $w \notin \mathrm{HP}$, then define $f(w) = 0 \notin O$.

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A problem:

- We can now compute whether a given input w is in HP by computing $f(w)$ and checking whether it is in O or not.
- But HP is not computable! Where is the issue?

Computable Functions

Definition (See Def. 2.1.17 in "Computability and Complexity" for full definition)

A function $f : \Sigma^* \to \Gamma^*$ is a computable function if and only if there exists a DTM M_f that on every input $w \in \Sigma^*$

- always halts
- with just $f(w)$ on its tape,
- and the head at the first letter of $f(w)$.

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Examples

- $f(w) = ww$ is computable.
- $f(\ulcorner m \urcorner \# \ulcorner n \urcorner) = \ulcorner m \cdot n \urcorner$ is computable (where $\ulcorner n \urcorner$ denotes the binary encoding of $n \in \mathbb{N}$).

A More Interesting Example

$$f(w) = \begin{cases} \ulcorner M' \urcorner & \text{if } w = \ulcorner M \urcorner \text{ for some Turing machine } M \\ & \text{and } M' \text{ is the Turing machine obtained} \\ & \text{from } M \text{ by replacing every occurrence of its} \\ & \text{rejecting state in a transition by the accepting one} \\ \varepsilon & \text{if } w \text{ is not an encoding of a Turing machine} \end{cases}$$

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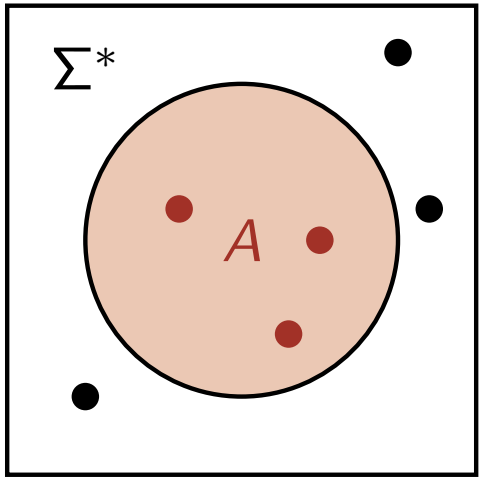
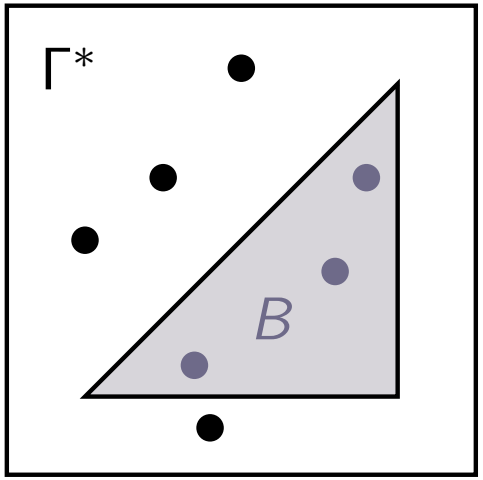
f is computed by the following Turing machine. On input w:

1. If w does not encode a Turing machine, empty the tape and accept.
2. Otherwise, replace every occurrence of r in w_δ by t (in the places where states are encoded).
3. Accept.

Mapping Reduction





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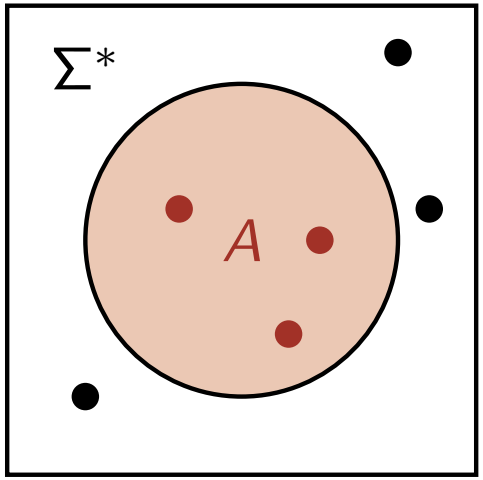

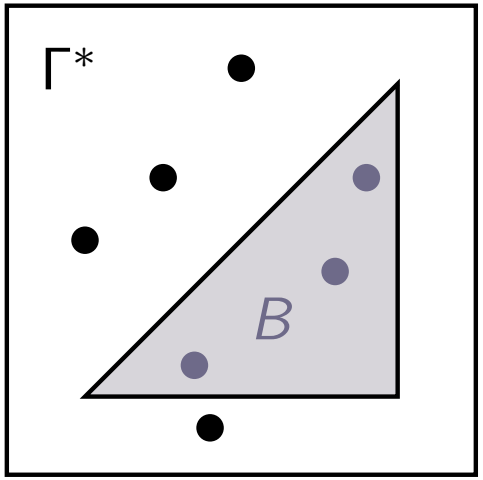

Definition

Let $A \subseteq \Sigma^*$ and $B \subseteq \Gamma^*$ be languages. We say that A is mapping reducible to B, written $A \leq_m B$, if and only if

1. there is a computable function $f\colon \Sigma^* \to \Gamma^*$ such that
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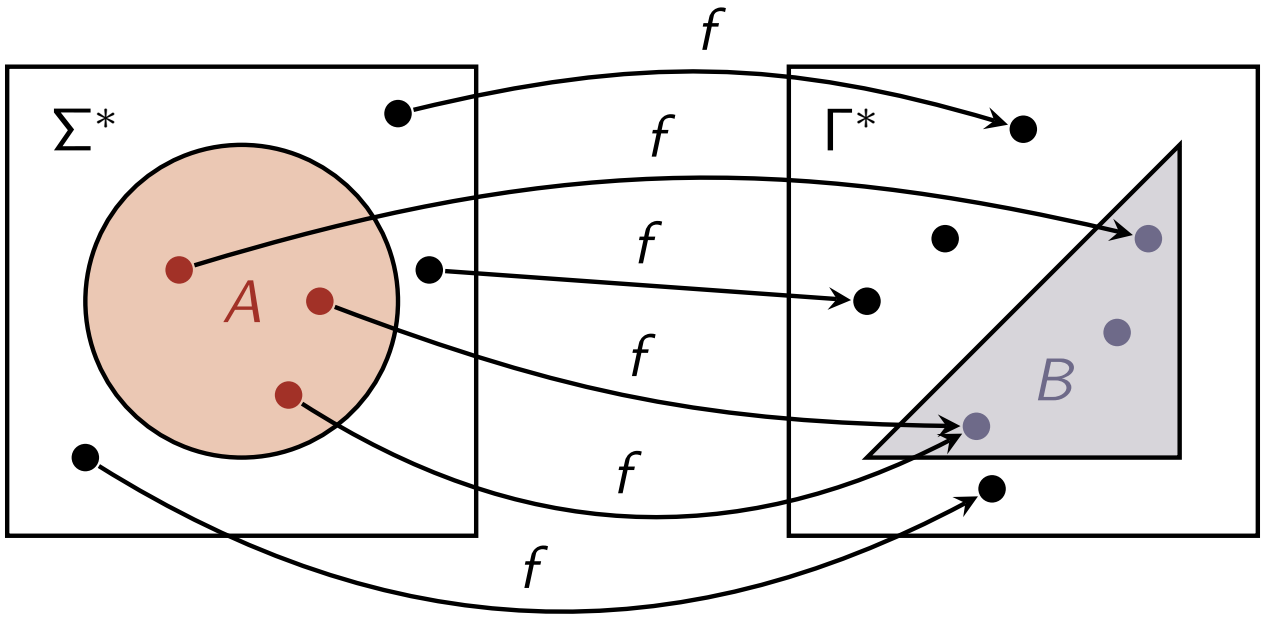

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Proof

- f is computable. So, g defined below is also computable:

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f is computable. We show $w \in \mathrm{AP} \Leftrightarrow f(w) \in \mathrm{HP}$:

- If $w \in \mathrm{AP}$, then $w = \langle \ulcorner M \urcorner, w' \rangle$ such that M accepts w'.
- Then, M' accepts w' as well, and in particular halts on w'.
- Hence, $f(w) = \langle \ulcorner M' \urcorner, w' \rangle \in \mathrm{HP}$.

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- Now assume $w \notin \mathrm{AP}$.
- If $f(w) = \varepsilon$ then $f(w) \notin \mathrm{HP}$.
- Otherwise, $w = \langle \ulcorner M \urcorner, w' \rangle$ such that M does not accept w'.
- Hence, either M rejects w' or M does not halt on w'.
- Let $f(\langle \ulcorner M \urcorner, w' \rangle) = \langle \ulcorner M' \urcorner, w' \rangle$.
- In both cases above, M' does not halt on w'.
- Thus, $f(w) = \langle \ulcorner M' \urcorner, w' \rangle \notin \mathrm{HP}$.

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- *If A is not computable, then B is not computable.*
- *If B is computably-enumerable, then A is computably-enumerable.*
- *If A is not computably-enumerable, then B is not computably-enumerable.*
- $\overline{A} \leq_m \overline{B}$.

Main Theorem

Theorem

Let $A \leq_m B$. Then:

- *If B is computable, then A is computable.*
- *If A is not computable, then B is not computable.*
- *If B is computably-enumerable, then A is computably-enumerable.*
- *If A is not computably-enumerable, then B is not computably-enumerable.*
- $\overline{A} \leq_m \overline{B}$.

Corollary

AP is not computable, but computably-enumerable.

Collatz

Recall Lecture 1: Does the following algorithm return `True` for every possible input $n \geq 1$?

```
def collatz(n):
  while(n > 1):
    if n%2 == 0:
      n = n/2
    else:
      n = 3*n+1
  return True
```

Collatz

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  return True
```

Note

If we could compute whether a DTM halts on every input, we could use this to solve the Collatz problem. However...

Non-computable Problems

The following problems are all non-computable:

- $\mathrm{HP} = \{\langle \ulcorner M \urcorner, w \rangle \mid M \text{ halts on input } w\}$
- $\mathrm{AP} = \{\langle \ulcorner M \urcorner, w \rangle \mid w \in L(M)\}$
- $\mathrm{NEP} = \{\ulcorner M \urcorner \mid L(M) \neq \emptyset\}$
- $\mathrm{UHP} = \{\ulcorner M \urcorner \mid M \text{ halts on every input}\}$
- $\mathrm{EQP} = \{\langle \ulcorner M \urcorner, \ulcorner M' \urcorner \rangle \mid L(M) = L(M')\}$
- The Entscheidungsproblem

There are many more in logics, automata theory, math, etc.

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There are many more in logics, automata theory, math, etc.

- Given six 3×3 integer matrices, can they be multiplied in some order (with possible repetitions) to yield the zero matrix?

Rice's Theorem

In a very specific sense, every interesting question about Turing machines is not computable!

Definition

Let $L \subseteq \Sigma^*$.

- L is nontrivial, if $L \neq \emptyset$ and $L \neq \Sigma^*$.
- L is semantic, if $\ulcorner M \urcorner \in L$ and $L(M) = L(M')$ implies $\ulcorner M' \urcorner \in L$ (membership of Turing machine encodings in L only depends on the accepted language).

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Examples

- $\{\ulcorner M \urcorner \mid M \text{ accepts } 09022024\}$ is semantic and nontrivial
- $\{\ulcorner M \urcorner \mid M \text{ accepts at least ten } w \text{ with } |w| > 11\}$ is semantic and nontrivial
- $\{\ulcorner M \urcorner \mid M \text{ has at least twenty states}\}$ is not semantic, but nontrivial

Rice's Theorem

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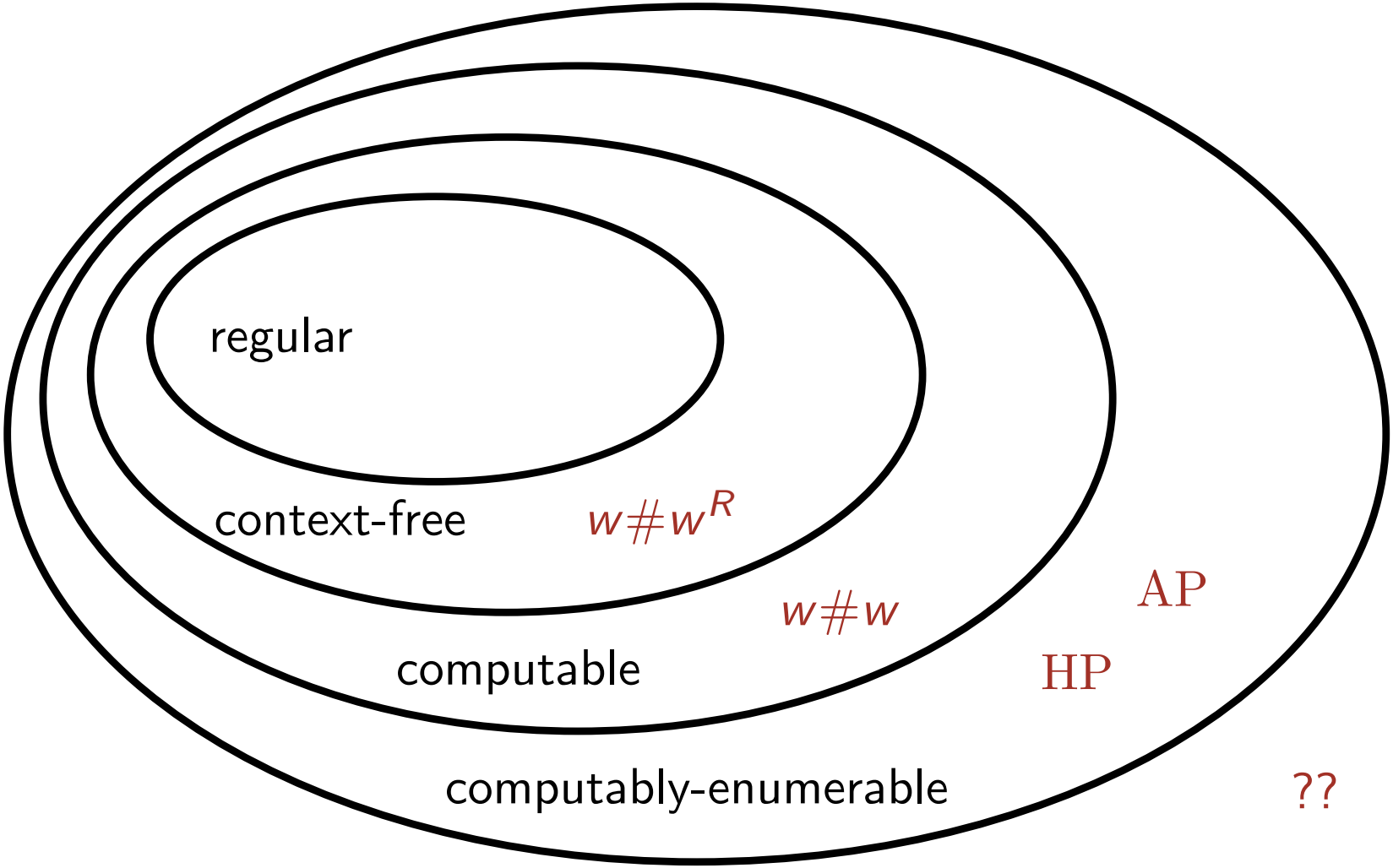
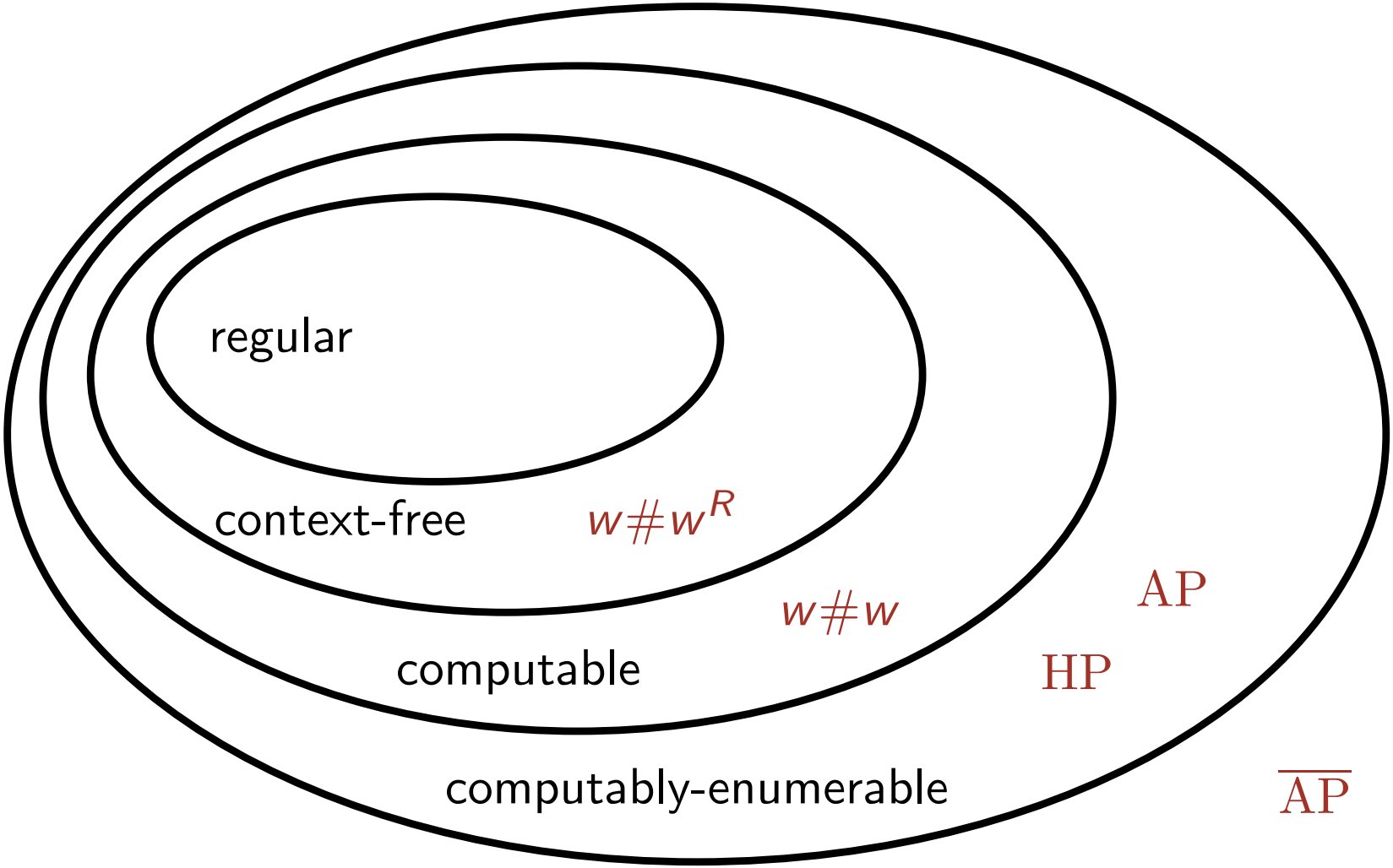
Theorem (Rice 1951)

Every nontrivial semantic language is non-computable.

Exercise


regular
context-free
$w\#w^R$
$w\#w$
computable
AP
HP
computably-enumerable
??

Exercise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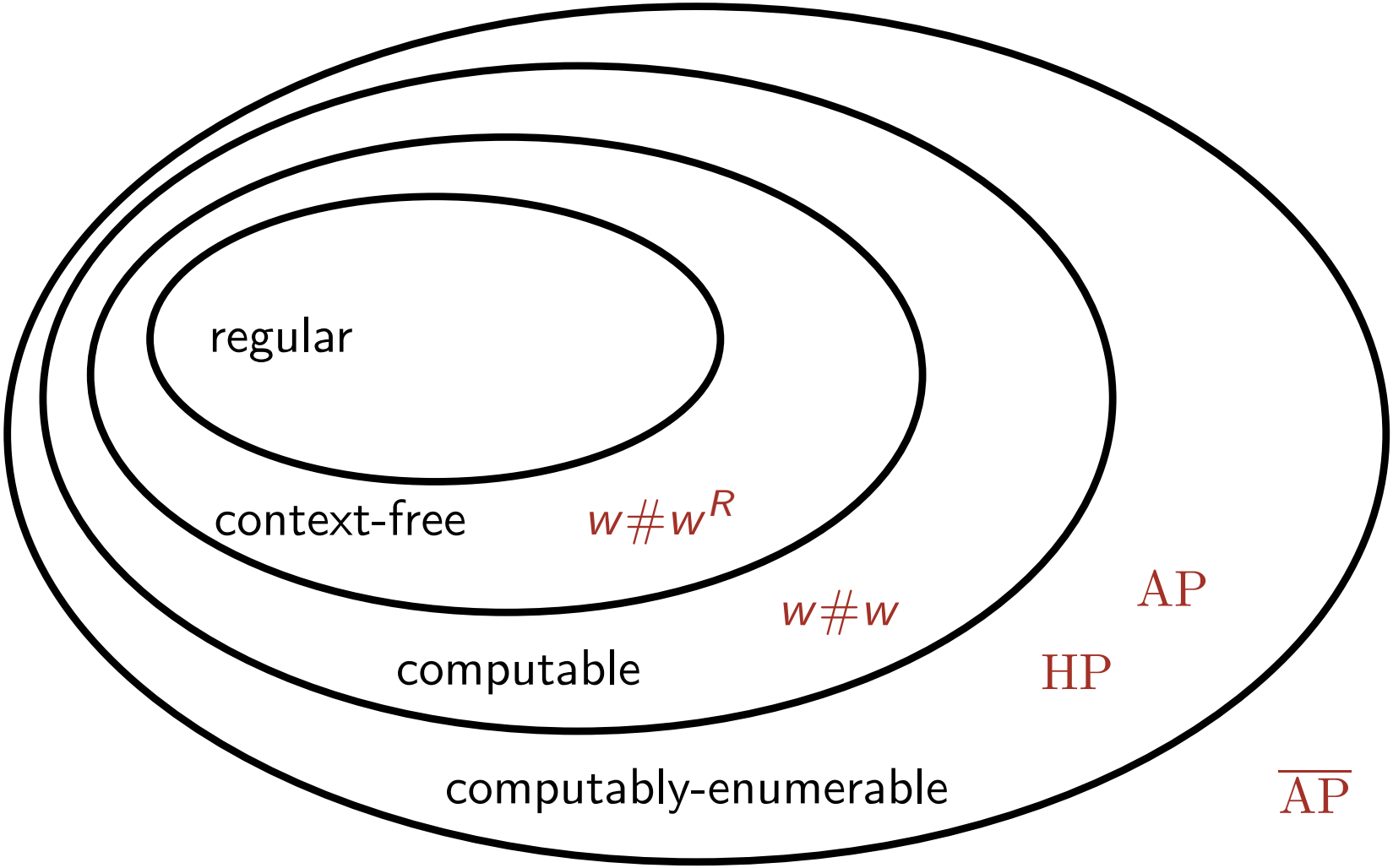

Note: AP is computably-enumerable. Thus, computably-enumerable languages are not closed under complementation (cf. Lecture 3).

Conclusion

We have seen

- Mapping reductions,
- more non-computable problems and how to prove them non-computable via reductions, and
- Rice's theorem: everything "interesting" about Turing machines is not computable.
- Consequence: everything "interesting" about programs is not computable.

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We have seen

- Mapping reductions,
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- Consequence: everything "interesting" about programs is not computable.

Reading

In "Computability and Complexity":

- Sections 2.4 and 2.8 on non-computable problems and reductions.
- Section 2.3 on universal Turing machines.
- Section 2.5 on closure properties.